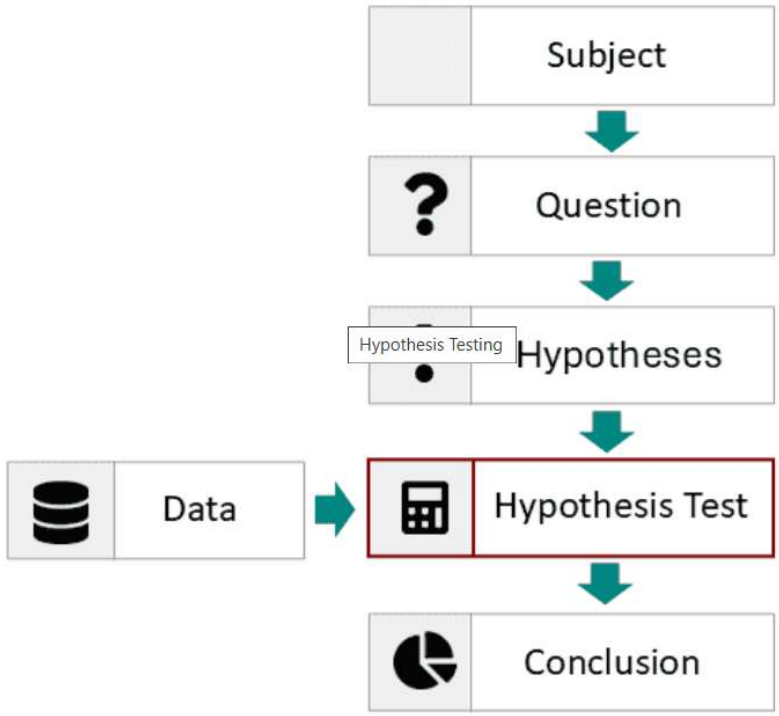
2201651-Expriment 7 Hypothesis Testing

1. What is need for Hypothesis Testing ?

A hypothesis test is used whenever you want to test a hypothesis about the population with the help of a sample. So whenever you want to prove or say something about the population with a sample, hypothesis tests are used.

Hypothesis testing is needed to make data-driven decisions. It helps determine whether a certain assumption about a population is likely to be true or false, based on sample data.
It provides a systematic and objective way to evaluate claims using statistics, minimizing bias and guesswork

2. With a neat diagram show important steps in hypothesis Testing.



Subject – Identify the topic or area of interest you want to study.

Question – Frame a research question or the problem you want to investigate.

Hypotheses – Formulate the null hypothesis (H_0) and alternative hypothesis (H_1) based on your question.

Data – Collect or use existing data relevant to your hypotheses.

Hypothesis Test – Perform a statistical test (like t-test, z-test, etc.) using the data to evaluate the hypotheses.

Conclusion – Based on the test result (p-value or test statistic), either accept or reject the null hypothesis, and draw a conclusion.

3. What is a student t distribution?

The Student's *t*-distribution is a probability distribution used in small sample scenarios (typically $n < 30$) when the population standard deviation is unknown.
It is wider than the normal distribution, allowing for more uncertainty due to fewer data points. It's used in t-tests for comparing sample means.

4. What is level of significance ? What role t plays in hypothesis Testing

A hypothesis test can never reject the null hypothesis with absolute certainty. There is always a certain probability of error that the null hypothesis is rejected even though it is actually true. This probability of error is called the significance level or α.

The significance level is used to decide whether the null hypothesis should be rejected or not. If the p-value is smaller than the significance level, the null hypothesis is to be rejected; otherwise, it is not to be rejected.

Usually, a significance level of 5% or 1% is set. If a significance level of 5% is set, it means that it is 5% likely to reject the null hypothesis even though it is actually true.

Role of t (t-statistic):

The t-statistic measures how far your sample mean is from the population/reference mean, in terms of standard error.

- In a hypothesis test, we calculate the t-statistic using the sample data.
- This value is then compared to the critical t-value from the t-distribution table (based on α and degrees of freedom).

Decision Rule:

- If |t-statistic| > t-critical, reject the null hypothesis (evidence of a difference).
- If |t-statistic| ≤ t-critical, fail to reject the null hypothesis (no strong evidence of a difference).

Hypotheses for Different t-Tests :

1. One-Sample t-test:

- H_0: Sample mean = Reference value
- H_1: Sample mean ≠ Reference value

2. Independent Samples t-test:

- H_0: Mean of Group A = Mean of Group B
- H_1: Mean of Group A ≠ Mean of Group B

3. Paired Sample t-test:

- H_0: Mean of differences = 0
- H_1: Mean of differences ≠ 0

5. Design a hypothesis testing for identifying average age of students in TE IT class is 19 or not.

We want to test if the average age of TE IT students is exactly 19 years.

Sample Data (ages):

ages = [18, 19, 20, 19, 18, 20, 21, 19, 18, 20]

Step 1: Define Hypotheses

- Null Hypothesis (H_0): $\mu = 19$ (Average age is 19)
- Alternative Hypothesis (H_1): $\mu \neq 19$ (Average age is not 19)

Step 2: Choose significance level

- $\alpha = 0.05$ (i.e., 95% confidence level)

Step 3: Choose the test

- Since we are comparing a sample mean with a population/reference value, and population standard deviation is unknown, we use a one-sample t-test.

Step 4: Decision Rule

- If the p-value < α → Reject H_0
- Else → Fail to reject H_0

6. Check if you can accept Ho at alpha 95% ? Match your conclusions with python functions.

We used a one-sample t-test to check if the average age is significantly different from 19.

```
from scipy import stats

# Sample data

ages = [18, 19, 20, 19, 18, 20, 21, 19, 18, 20]

# Perform one-sample t-test

t_statistic, p_value = stats.ttest_1samp(ages, 19)

print(f"t-statistic: {t_statistic:.4f}")

print(f"p-value: {p_value:.4f}")
```

OUTPUT:

t-statistic: 0.6124

p-value: 0.5554

```python
from scipy import stats

# Sample data
ages = [18, 19, 20, 19, 18, 20, 21, 19, 18, 20]

# Perform one-sample t-test
t_statistic, p_value = stats.ttest_1samp(ages, 19)

print(f"t-statistic: {t_statistic:.4f}")
print(f"p-value: {p_value:.4f}")
```

```
t-statistic: 0.6124
p-value: 0.5554
```

Decision Rule:

- Significance level α = 0.05
- Since p-value = 0.5554 > 0.05, we fail to reject H_0.

Conclusion:
There is no significant difference between the sample mean and 19.
Hence, we accept the null hypothesis (H_0) — the average age of TE IT students is 19.